

1. Administrative & Study Identification

Full Study Title: Study to Assess Safety, Efficacy, Pharmacokinetics, and Pharmacodynamics of Crovalimab in Healthy Volunteers and Participants With Paroxysmal Nocturnal Hemoglobinuria **Short Title/Acronym:** COMPOSER **Protocol Number:** BP39144 **NCT Number:** NCT03157635 **Sponsor Name:** Roche **Investigational Product:** Crovalimab (SKY59 / R07112689) **Phase of Development:** Phase I/II **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, pharmacokinetics (PK), and pharmacodynamics (PD) of single and multiple doses of crovalimab in healthy volunteers and participants with PNH. **Secondary Objectives:** To evaluate efficacy endpoints including change in lactate dehydrogenase (LDH) levels, terminal complement activity, transfusion avoidance, hemoglobin stabilization, and breakthrough hemolysis (BTH).

2.2 Background & Rationale Complement C5 inhibition is the standard of care for patients with PNH who have significant clinical symptoms. Complete and constant suppression of the terminal complement pathway is necessary, which traditionally required frequent intravenous treatments. Crovalimab is a novel anti-C5 monoclonal antibody with sequential recycling technology engineered for extended, self-administered subcutaneous dosing of small volumes. This study aims to evaluate if crovalimab can provide complete terminal complement inhibition while significantly reducing the treatment administration burden for patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (Part 1 for healthy volunteers) and Active/Uncontrolled (Parts 2-4 for PNH patients) **Blinding:** Double-blind (Part 1) and Open-label (Parts 2-4 and Extension) **Randomization:** Randomized (Part 1 & 3) and Non-randomized (Part 2 & 4)

3.2 Planned Enrollment & Duration Target **SN\$:** 59 participants (healthy volunteers and PNH patients) **Number of Sites:** Global multicenter **Estimated Study Start:** 14-Nov-2016 **Estimated Primary Completion:** 30-Sep-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy male volunteers aged 21-55 years (Part 1). 2. Documented diagnosis of PNH with LDH $\geq 1.5\times$ ULN (if treatment-naïve) or previously treated continuously with eculizumab for ≥ 3 months. 3. Vaccinated against *Neisseria meningitidis* serogroups A, C, W, Y, and B.

4.2 Exclusion Criteria 1. Any clinically relevant history or presence of moderate to severe respiratory, renal, hepatic, gastrointestinal, hematological, lymphatic, neurological, or cardiovascular disease. 2. Prior splenectomy ≤ 1 year before starting crovalimab. 3. Active infections including HIV, chronic active hepatitis C, or a history of meningococcal meningitis.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Loading Dose: Intravenous (IV) dose on Day 1 (1000 mg for < 100 kg; 1500 mg for ≥ 100 kg), followed by 340 mg subcutaneous (SC) on Days 2, 8, 15, and 22. * Maintenance Dose: 680 mg (< 100 kg) or 1020 mg (≥ 100 kg) administered SC every 4 weeks (Q4W) starting Week 5. **Route of Administration:** Intravenous (IV) loading and Subcutaneous (SC) maintenance. **Duration of Treatment:** Sequential study parts followed by an Open-Label Extension (OLE) running until study completion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive therapies for PNH. **Prohibited:** Treatment with azathioprine or erythrocyte-stimulating agents within 14 days prior to first study drug administration, or active therapy with intravenous immunoglobulin (IVIG).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Vaccination verification, Labs (LDH)
Baseline	Day 1	First IV Dose, PK/PD parameters, Safety baseline
Interim	Days 2, 8, 15, 22; Week 5 and Q4W	SC Loading doses; Maintenance SC doses, LDH assessments, Safety and Efficacy monitoring
End of Study	Up to 2026	Final safety and efficacy evaluations (OLE completion)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: * **Part 1:** Percentage of Participants With Dose-Limiting Events (DLEs). * **Part 1:**

Percentage of Participants With Adverse Events (AEs) and Serious Adverse Events (SAEs). * **Part 2:** Percentage of Participants With AEs and SAEs. **Measurement Method:** Clinical evaluations, safety reports, and continuous laboratory monitoring.

7.2 Secondary & Exploratory Endpoints * **Part 1:** Terminal Complement Activity as Assessed by Ex Vivo Liposome Immunoassay (LIA). * **Part 2:** Serum Lactate Dehydrogenase (LDH) Levels. * **Part 3:** Serum Lactate Dehydrogenase (LDH) Levels. * Exploratory endpoints: Transfusion avoidance, hemoglobin stabilization, and rates of breakthrough hemolysis (BTH). Terminal complement pathway inhibition defined as free C5 < 50 ng/mL.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard monitoring via clinical visits and laboratory assessments, specifically observing for drug-target-drug complexes (e.g., transient vasculitic skin reactions when switching from eculizumab). **Serious Adverse Events (SAEs):** Reported within 24 hours (including monitoring for severe infections such as meningococcal disease). **Data Monitoring Committee (DMC):** Safety review committee setup per Roche's internal and external guidelines.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy parameters; Safety Set for all AEs. **Sample Size Justification:** Designed as an adaptive Phase I/II study focused on establishing safety, PK/PD parameters, and initial efficacy profiles, rather than powered for formal hypothesis testing. **Handling of Missing Data:** Standard imputation methodologies (e.g., LOCF) for continuous biochemical variables such as LDH.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Onsite and remote monitoring to ensure protocol adherence and safety. **Audit Trail:** Electronic Case Report Form (eCRF) locking and standard data clarification processes.

11. Study Identifiers & Governance

Primary ID: BP39144 **Registry Number:** NCT03157635 **Posting ID:** 1133225479557152908 **Sponsor/Lead Organization:** Roche **Study Title:** COMPOSER **Official Regulatory Title:** An Adaptive Phase I/II Study to Assess Safety, Efficacy, Pharmacokinetics and Pharmacodynamics

of Crovalimab in Healthy Volunteers and Patients With Paroxysmal Nocturnal Hemoglobinuria (PNH)

12. Clinical Framework (Paroxysmal Nocturnal Hemoglobinuria Specific)

Primary Condition: Paroxysmal Nocturnal Hemoglobinuria (PNH) / Paroxysmal Hemoglobinuria, Nocturnal **Disease Semantic Type:** Disease or Syndrome **Medical Terminologies:** MeSH: D006457 | HP0: HP:0004818 | SNOMED: 1963002 **Diagnosis Threshold:** LDH $\geq 1.5\times$ ULN and/or PNH granulocyte clone size $\geq 10\%$ **Medical Methodology:** Terminal Complement C5 inhibition **Optimization Target:** Complete and sustained terminal complement pathway inhibition (free C5 $< 50\text{ ng/mL}$)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for PNH (Treatment-naive or switching from eculizumab) **Education Protocol (HCP):** Training on Subcutaneous (SC) dosing protocols and mandatory meningococcal vaccination verification **Education Protocol (Patient):** SC self-administration technique and infection signs awareness **Quality Audit Mechanism:** HCP perception tracking & patient adherence monitoring to the Q4W SC regimen

14. Observation & Follow-Up Schedule

Total Duration: Follow-up through Open-Label Extension (OLE) completion (estimated up to ~10 years) **Onsite Visit Cadence:** Day 1, 8, 15, 22, Week 5, and Q4W thereafter **Remote Monitoring Frequency:** As required between maintenance doses **Checklist Review Interval:** Every 12 weeks (Q12W)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				